No.	Status	Date
HIS-2006-0412-DS-2-RV-102	Issued	2006-02-24

UT Thickness Data Sheet - 2-RV-102

Scope and Responsibilities

This data sheet provides the ultrasonic thickness measurement results for line 2-RV-102, located within the ATU-100. The inspection was conducted as part of the 10-year full-unit survey detailed in HIS-2006-0412.

Kevin Doucette, acting as the lead UT technician (SNT-TC-1A UT Level II) for Halward Inspection Services Ltd., performed the measurements and data recording for this specific line.

Technical Basis and Procedures

Thickness measurements were performed in accordance with the site NDE procedure QSCP-NDT-001 Rev 0 and the Halward Inspection Services Ltd. procedure HIS-UT-002 Rev 1.

The inspection methodology and equipment calibration were executed in compliance with the requirements of ASTM E797.

Component Design Data

The subject line, 2-RV-102, is the regenerator overhead constructed from A106 Gr.B. The design specifications are an outer diameter of 60.3 mm with a nominal wall thickness of 5.54 mm.

The design limits for this component are 345 kPa(g) and 121 °C. The minimum allowable thickness (t-min) is 2.79 mm.

Assessment Criteria

Acceptance is based on the retirement thickness, which is defined as the t-min of 2.79 mm. Any reading below this value requires the issuance of an NCR.

Remaining life and the interval for the next thickness reading are determined by the corrosion rate, calculated as the larger of the long-term rate or the short-term rate over the latest pair of readings at least 3 years apart.

Inspection Interval

Line 2-RV-102 is categorized as Class 1 due to its service as regenerator overhead in sour gas. The maximum interval for thickness measurements and external visual inspection is 5 years.

The next inspection date shall be determined by the remaining life calculation, not to exceed the Class 1 maximum of 5 years.

Thickness readings, measured 2006-02-24

Line 6-SG-101 | Sour gas inlet | A106 Gr.B | t_nom 7.11 mm | t-min 3.56 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
6-SG-101-01	7.11	3.56	2001-03-02	6.82	6.59	0.053	0.046	>50	2011-02-24	
6-SG-101-02	7.11	3.56	2001-03-02	6.71	6.51	0.054	0.040	>50	2011-02-24	
6-SG-101-03	7.11	3.56	2001-03-02	6.74	6.42	0.066	0.064	43.2	2011-02-24	
6-SG-101-04	7.11	3.56	2001-03-02	6.73	6.28	0.081	0.090	30.1	2011-02-24	
6-SG-101-05	7.11	3.56	2001-03-02	6.68	6.22	0.092	0.092	28.8	2011-02-24	
6-SG-101-06	7.11	3.56	2001-03-02	6.87	6.49	0.058	0.076	38.4	2011-02-24	

Line 6-SG-102 | Sour gas inlet | A106 Gr.B | t_nom 7.11 mm | t-min 3.56 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
6-SG-102-01	7.11	3.56	2001-03-02	6.72	6.26	0.088	0.092	29.2	2011-02-24	
6-SG-102-02	7.11	3.56	2001-03-02	6.72	6.25	0.086	0.094	28.5	2011-02-24	
6-SG-102-03	7.11	3.56	2001-03-02	6.69	6.26	0.078	0.086	31.3	2011-02-24	
6-SG-102-04	7.11	3.56	2001-03-02	6.64	6.22	0.077	0.084	31.6	2011-02-24	
6-SG-102-05	7.11	3.56	2001-03-02	6.93	6.72	0.041	0.042	>50	2011-02-24	

Line 8-TG-101 | Treated gas | A106 Gr.B | t_nom 8.18 mm | t-min 4.06 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
8-TG-101-01	8.18	4.06	1996-03-08	8.20	7.89	0.031	0.031	>50	2016-02-24	
8-TG-101-02	8.18	4.06	1996-03-08	8.20	7.75	0.045	0.045	>50	2016-02-24	
8-TG-101-03	8.18	4.06	1996-03-08	8.23	7.79	0.044	0.044	>50	2016-02-24	
8-TG-101-04	8.18	4.06	1996-03-08	8.10	7.76	0.034	0.034	>50	2016-02-24	
8-TG-101-05	8.18	4.06	1996-03-08	8.22	7.95	0.027	0.027	>50	2016-02-24	

Line 4-LA-101 | Lean amine supply | A106 Gr.B | t_nom 6.02 mm | t-min 3.18 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
4-LA-101-01	6.02	3.18	1996-03-08	5.99	5.51	0.048	0.048	48.4	2016-02-24	
4-LA-101-02	6.02	3.18	1996-03-08	6.01	5.76	0.025	0.025	>50	2016-02-24	
4-LA-101-03	6.02	3.18	1996-03-08	5.96	5.76	0.020	0.020	>50	2016-02-24	
4-LA-101-04	6.02	3.18	1996-03-08	6.07	5.78	0.029	0.029	>50	2016-02-24	
4-LA-101-05	6.02	3.18	1996-03-08	6.01	5.81	0.020	0.020	>50	2016-02-24	
4-LA-101-06	6.02	3.18	1996-03-08	6.06	5.64	0.042	0.042	>50	2016-02-24	
4-LA-101-07	6.02	3.18	1996-03-08	5.99	5.57	0.042	0.042	>50	2016-02-24	

Line 4-LA-102 | Lean amine supply | A106 Gr.B | t_nom 6.02 mm | t-min 3.18 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
4-LA-102-01	6.02	3.18	1996-03-08	6.02	5.63	0.039	0.039	>50	2016-02-24	
4-LA-102-02	6.02	3.18	1996-03-08	6.07	5.67	0.040	0.040	>50	2016-02-24	
4-LA-102-03	6.02	3.18	1996-03-08	6.07	5.55	0.052	0.052	45.4	2016-02-24	
4-LA-102-04	6.02	3.18	1996-03-08	5.99	5.34	0.065	0.065	33.1	2016-02-24	
4-LA-102-05	6.02	3.18	1996-03-08	6.00	5.62	0.038	0.038	>50	2016-02-24	
4-LA-102-06	6.02	3.18	1996-03-08	6.00	5.83	0.017	0.017	>50	2016-02-24	
4-LA-102-07	6.02	3.18	1996-03-08	6.00	5.63	0.037	0.037	>50	2016-02-24	

Line 6-LA-103 | Lean amine header | A106 Gr.B | t_nom 7.11 mm | t-min 3.56 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
6-LA-103-01	7.11	3.56	1996-03-08	7.02	6.75	0.027	0.027	>50	2016-02-24	
6-LA-103-02	7.11	3.56	1996-03-08	7.19	6.87	0.032	0.032	>50	2016-02-24	
6-LA-103-03	7.11	3.56	1996-03-08	7.08	6.65	0.043	0.043	>50	2016-02-24	
6-LA-103-04	7.11	3.56	1996-03-08	7.17	6.52	0.065	0.065	45.4	2016-02-24	
6-LA-103-05	7.11	3.56	1996-03-08	7.19	6.49	0.070	0.070	41.7	2016-02-24	
6-LA-103-06	7.11	3.56	1996-03-08	7.16	6.42	0.074	0.074	38.5	2016-02-24	
6-LA-103-07	7.11	3.56	1996-03-08	7.18	6.76	0.042	0.042	>50	2016-02-24	
6-LA-103-08	7.11	3.56	1996-03-08	7.04	6.79	0.025	0.025	>50	2016-02-24	

Line 4-RA-101 | Rich amine | A106 Gr.B | t_nom 8.56 mm | t-min 3.81 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
4-RA-101-01	8.56	3.81	2001-03-02	8.39	8.27	0.025	0.024	>50	2016-02-24	
4-RA-101-02	8.56	3.81	2001-03-02	8.45	8.27	0.027	0.036	>50	2016-02-24	
4-RA-101-03	8.56	3.81	2001-03-02	8.42	8.10	0.051	0.064	>50	2016-02-24	
4-RA-101-04	8.56	3.81	2001-03-02	8.37	8.34	0.029	0.006	>50	2016-02-24	
4-RA-101-05	8.56	3.81	2001-03-02	8.42	8.29	0.031	0.026	>50	2016-02-24	
4-RA-101-07	8.56	3.81	2001-03-02	8.51	8.30	0.030	0.042	>50	2016-02-24	
4-RA-101-08	8.56	3.81	2001-03-02	8.30	8.07	0.061	0.046	>50	2016-02-24	
4-RA-101-09	8.56	3.81	2001-03-02	8.43	8.43	0.014	0.000	>50	2016-02-24	
4-RA-101-10	8.56	3.81	2001-03-02	8.48	8.27	0.040	0.042	>50	2016-02-24	
4-RA-101-11	8.56	3.81	2001-03-02	8.39	8.14	0.054	0.050	>50	2016-02-24	

Line 4-RA-102 | Rich amine | A106 Gr.B | t_nom 8.56 mm | t-min 3.81 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
4-RA-102-01	8.56	3.81	2001-03-02	8.48	8.41	0.013	0.014	>50	2016-02-24	
4-RA-102-02	8.56	3.81	2001-03-02	8.50	8.15	0.042	0.070	>50	2016-02-24	
4-RA-102-03	8.56	3.81	2001-03-02	8.37	8.27	0.029	0.020	>50	2016-02-24	
4-RA-102-04	8.56	3.81	2001-03-02	8.35	8.24	0.030	0.022	>50	2016-02-24	
4-RA-102-05	8.56	3.81	2001-03-02	8.36	8.19	0.036	0.034	>50	2016-02-24	
4-RA-102-06	8.56	3.81	2001-03-02	8.38	8.15	0.036	0.046	>50	2016-02-24	
4-RA-102-07	8.56	3.81	2001-03-02	8.49	8.29	0.029	0.040	>50	2016-02-24	
4-RA-102-08	8.56	3.81	2001-03-02	8.36	8.05	0.044	0.062	>50	2016-02-24	
4-RA-102-09	8.56	3.81	2001-03-02	8.35	8.18	0.037	0.034	>50	2016-02-24	

Line 6-RA-103 | Rich amine header | A106 Gr.B | t_nom 10.97 mm | t-min 4.57 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
6-RA-103-01	10.97	4.57	2001-03-02	10.82	10.76	0.024	0.012	>50	2016-02-24	
6-RA-103-02	10.97	4.57	2001-03-02	10.69	10.50	0.059	0.038	>50	2016-02-24	
6-RA-103-03	10.97	4.57	2001-03-02	10.84	10.67	0.032	0.034	>50	2016-02-24	
6-RA-103-04	10.97	4.57	2001-03-02	10.59	10.40	0.058	0.038	>50	2016-02-24	
6-RA-103-05	10.97	4.57	2001-03-02	10.61	10.34	0.061	0.054	>50	2016-02-24	
6-RA-103-06	10.97	4.57	2001-03-02	10.86	10.74	0.027	0.024	>50	2016-02-24	
6-RA-103-07	10.97	4.57	2001-03-02	10.79	10.66	0.033	0.026	>50	2016-02-24	
6-RA-103-08	10.97	4.57	2001-03-02	10.83	10.58	0.039	0.050	>50	2016-02-24	
6-RA-103-09	10.97	4.57	2001-03-02	10.85	10.67	0.034	0.036	>50	2016-02-24	
6-RA-103-10	10.97	4.57	2001-03-02	10.78	10.27	0.075	0.102	>50	2016-02-24	

Line 3-RV-101 | Regenerator overhead | A106 Gr.B | t_nom 7.62 mm | t-min 3.56 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
3-RV-101-01	7.62	3.56	2001-03-02	7.26	6.98	0.061	0.056	>50	2011-02-24	
3-RV-101-02	7.62	3.56	2001-03-02	7.31	6.93	0.064	0.076	44.2	2011-02-24	
3-RV-101-03	7.62	3.56	2001-03-02	7.15	6.57	0.095	0.116	25.9	2011-02-24	
3-RV-101-04	7.62	3.56	2001-03-02	7.18	6.65	0.094	0.106	29.1	2011-02-24	
3-RV-101-05	7.62	3.56	2001-03-02	7.25	7.01	0.066	0.048	>50	2011-02-24	
3-RV-101-06	7.62	3.56	2001-03-02	7.20	6.74	0.085	0.092	34.4	2011-02-24	

Line 2-RV-102 | Regenerator overhead | A106 Gr.B | t_nom 5.54 mm | t-min 2.79 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
2-RV-102-01	5.54	2.79	2001-03-02	5.29	4.85	0.068	0.088	23.3	2011-02-24	
2-RV-102-02	5.54	2.79	2001-03-02	5.12	4.84	0.075	0.056	27.2	2011-02-24	
2-RV-102-03	5.54	2.79	2001-03-02	5.24	5.14	0.034	0.020	>50	2011-02-24	
2-RV-102-04	5.54	2.79	2001-03-02	5.33	5.07	0.051	0.052	43.7	2011-02-24	

Line 3-RF-101 | Reflux | A312 TP316 | t_nom 5.49 mm | t-min 2.41 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
3-RF-101-01	5.49	2.41	1996-03-08	5.48	5.48	0.000	0.000	>50	2016-02-24	
3-RF-101-02	5.49	2.41	1996-03-08	5.45	5.37	0.008	0.008	>50	2016-02-24	
3-RF-101-03	5.49	2.41	1996-03-08	5.45	5.45	0.000	0.000	>50	2016-02-24	
3-RF-101-04	5.49	2.41	1996-03-08	5.46	5.33	0.013	0.013	>50	2016-02-24	
3-RF-101-05	5.49	2.41	1996-03-08	5.45	5.40	0.005	0.005	>50	2016-02-24	
3-RF-101-06	5.49	2.41	1996-03-08	5.52	5.40	0.012	0.012	>50	2016-02-24	
3-RF-101-07	5.49	2.41	1996-03-08	5.42	5.35	0.007	0.007	>50	2016-02-24	

Line 2-RF-102 | Reflux | A312 TP316 | t_nom 3.91 mm | t-min 2.16 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
2-RF-102-01	3.91	2.16	1996-03-08	3.92	3.85	0.007	0.007	>50	2016-02-24	
2-RF-102-02	3.91	2.16	1996-03-08	3.88	3.88	0.000	0.000	>50	2016-02-24	
2-RF-102-03	3.91	2.16	1996-03-08	3.93	3.80	0.013	0.013	>50	2016-02-24	
2-RF-102-04	3.91	2.16	1996-03-08	3.87	3.81	0.006	0.006	>50	2016-02-24	
2-RF-102-07	3.91	2.16	1996-03-08	3.96	3.85	0.011	0.011	>50	2016-02-24	

Line 4-LA-104 | L/R exchanger, lean side | A106 Gr.B | t_nom 6.02 mm | t-min 3.18 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
4-LA-104-01	6.02	3.18	1996-03-08	6.07	5.75	0.032	0.032	>50	2016-02-24	
4-LA-104-02	6.02	3.18	1996-03-08	6.03	5.56	0.047	0.047	>50	2016-02-24	
4-LA-104-03	6.02	3.18	1996-03-08	5.96	5.74	0.022	0.022	>50	2016-02-24	
4-LA-104-04	6.02	3.18	1996-03-08	6.06	5.86	0.020	0.020	>50	2016-02-24	
4-LA-104-05	6.02	3.18	1996-03-08	6.04	5.51	0.053	0.053	43.8	2016-02-24	
4-LA-104-06	6.02	3.18	1996-03-08	6.10	5.83	0.027	0.027	>50	2016-02-24	
4-LA-104-07	6.02	3.18	1996-03-08	6.01	5.62	0.039	0.039	>50	2016-02-24	
4-LA-104-08	6.02	3.18	1996-03-08	5.94	5.50	0.044	0.044	>50	2016-02-24	

Line 4-RA-104 | L/R exchanger, rich side | A106 Gr.B | t_nom 8.56 mm | t-min 3.81 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
4-RA-104-01	8.56	3.81	2001-03-02	8.40	8.16	0.035	0.048	>50	2016-02-24	
4-RA-104-02	8.56	3.81	2001-03-02	8.41	8.23	0.032	0.036	>50	2016-02-24	
4-RA-104-03	8.56	3.81	2001-03-02	8.46	8.30	0.024	0.032	>50	2016-02-24	
4-RA-104-04	8.56	3.81	2001-03-02	8.35	8.21	0.038	0.028	>50	2016-02-24	
4-RA-104-05	8.56	3.81	2001-03-02	8.29	8.14	0.046	0.030	>50	2016-02-24	
4-RA-104-06	8.56	3.81	2001-03-02	8.39	8.09	0.050	0.060	>50	2016-02-24	
4-RA-104-07	8.56	3.81	2001-03-02	8.29	7.97	0.062	0.064	>50	2016-02-24	

Line 2-ST-101 | Reboiler steam | A106 Gr.B | t_nom 3.91 mm | t-min 2.29 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
2-ST-101-01	3.91	2.29	1996-03-08	3.93	3.64	0.029	0.029	46.4	2016-02-24	
2-ST-101-02	3.91	2.29	1996-03-08	3.90	3.67	0.023	0.023	>50	2016-02-24	
2-ST-101-03	3.91	2.29	1996-03-08	3.94	3.82	0.012	0.012	>50	2016-02-24	
2-ST-101-04	3.91	2.29	1996-03-08	3.82	3.77	0.005	0.005	>50	2016-02-24	
2-ST-101-05	3.91	2.29	1996-03-08	3.91	3.72	0.019	0.019	>50	2016-02-24	

Line 3-CW-101 | Amine cooler CW | A106 Gr.B | t_nom 5.49 mm | t-min 2.79 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
3-CW-101-01	5.49	2.79	1996-03-08	5.42	5.13	0.029	0.029	>50	2016-02-24	
3-CW-101-02	5.49	2.79	1996-03-08	5.41	5.20	0.021	0.021	>50	2016-02-24	
3-CW-101-03	5.49	2.79	1996-03-08	5.53	5.08	0.045	0.045	>50	2016-02-24	
3-CW-101-04	5.49	2.79	1996-03-08	5.46	5.38	0.008	0.008	>50	2016-02-24	
3-CW-101-05	5.49	2.79	1996-03-08	5.48	5.33	0.015	0.015	>50	2016-02-24	

Line 3-CW-102 | Amine cooler CW | A106 Gr.B | t_nom 5.49 mm | t-min 2.79 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
3-CW-102-01	5.49	2.79	1996-03-08	5.50	5.20	0.030	0.030	>50	2016-02-24	
3-CW-102-02	5.49	2.79	1996-03-08	5.51	5.02	0.049	0.049	45.4	2016-02-24	
3-CW-102-03	5.49	2.79	1996-03-08	5.53	5.07	0.046	0.046	49.4	2016-02-24	
3-CW-102-04	5.49	2.79	1996-03-08	5.51	5.03	0.048	0.048	46.5	2016-02-24	

Sign-off	Name	Function	Date	Signature
Prepared	Kevin Doucette	lead UT technician, SNT-TC-1A UT Level II	2006-02-24	_____